

Prediction–Attribution Conjugacy in Adaptive Decision Problems

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October 2026

Abstract

Many consequential decisions require both a forecast and an account of which parts or relations within a system matter for action. We examine adaptive decision problems in which decision makers consider structural accounts, make prospective predictions, act, and revise their understanding as outcomes become available. Prediction and attribution become diagnostically reciprocal when competing accounts imply different outcomes under the same informative condition, allowing later observations to change their relative credibility. We call this relationship prediction–attribution conjugacy. The framework distinguishes that diagnostic core from learning and from the system’s response to action. These distinctions help explain why routine predictive stability may leave attribution unresolved, why successful intervention can weaken an old predictive signal, and why diagnostic learning may be overtaken by structural change. Examples from finance, clinical trials, and public policy show how the usefulness of the framework depends on the decision, the available diagnostic comparisons, and what action changes.

Keywords: prediction; attribution; adaptive decision problems; decision making; model monitoring; identification.

1. Introduction

A portfolio risk manager expects the S&P 500 to remain heavily concentrated in a small number of large technology companies. The forecast matters, but it does not by itself tell the manager what to do.

An effective response depends on what lies behind the concentration. If the concern is market-cap weighting, one strategy may be appropriate. If several large companies share exposure to the same underlying economic factor, another may be needed. If the concentration instead reflects persistent differences in earnings growth among businesses with distinct risks, mechanically reducing the largest positions may create a different problem. The manager needs an account of which relationships make the concentration consequential for this portfolio.

That account also changes what the manager should predict. A forecast that concentration will persist may prompt a more useful question: how would the portfolio perform under a shock to a shared exposure? Once that outcome and shock are specified, competing accounts may predict different losses. Observing the portfolio under a sufficiently informative condition could then help determine which account remains credible.

This combination appears in other consequential decisions. A physician may need to anticipate deterioration and understand enough about the source of risk to choose a treatment. A policymaker may need to predict an intervention’s effects and understand why similar interventions worked differently across populations or periods. An investor may find a profitable regularity but need an account of its

source to judge whether the opportunity will survive exploitation. In each case, the decision gives understanding a particular task: identify the distinctions that could change what should be done.

We examine adaptive decision problems in which uncertainty concerns both the outcome and the structure relevant to action. Decision makers consider alternative accounts, act on the understanding available, observe subsequent outcomes, and may revise their accounts or choices. The relevant system may also change during that process. This defines the setting for our inquiry. Whether the available evidence can distinguish the accounts is a further question.

Within some of these problems, attribution and prediction have a reciprocal diagnostic relationship. An attribution implies what should happen under a specified condition, and what subsequently happens can change the evidentiary balance among competing attributions. We call this *prediction-attribution conjugacy*. The term names a defined coupling between two distinct inquiries. It carries no presumption that they are mathematical inverses or that improving one must worsen the other.

The relation becomes especially consequential when understanding supports action in a responsive system. A successful intervention may change the distribution or relationship that supplied an earlier predictive signal. A subsequent change in predictive performance can therefore require several explanations: the original account may have been mistaken, the relevant structure may have changed independently, or action may have successfully changed what the predictor relied on. Distinguishing those explanations is part of understanding whether the decision worked.

Our purpose is to organize these questions around the decision maker's need for usable understanding. Section 2 establishes the decision-relevant system. Sections 3 and 4 distinguish attribution-guided formulation from the diagnostic relation itself. Section 5 follows that relation into learning and responsive action, and Section 6 uses contrasting cases to establish its boundaries. Sections 7 and 8 connect the account to existing research and monitoring practice. Appendix A supplies the notation and a worked comparison; Appendices B and C develop the classification and exploratory extensions.

2. The Adaptive Decision Problem

We begin with the decision because its objective determines which parts of an uncertain system deserve attention. A decision maker generally needs enough understanding to choose among relevant actions, rather than a complete account of the wider system.

We call the portion or representation of the system that matters for the objective the *decision-relevant system*, denoted S_D . The subscript D indicates that the representation is organized around a particular decision objective. Appendix A.1 formalizes this relationship.

Consider again the stock market. A manager concerned with concentration risk and a hedge fund manager seeking arbitrage opportunities may observe the same S&P 500. The risk manager may organize the evidence around common exposures, drawdowns, liquidity, correlations, and contributions to portfolio loss. The arbitrageur may instead attend to mispricing, convergence mechanisms, execution costs, crowding, and the likely persistence of an opportunity. These representations answer different decision questions.

This gives attribution a more definite object. “Why did the market move?” admits a wide range of explanations. “Which relationships account for the losses this portfolio is trying to manage?” directs inquiry toward the components and conditions that could change a risk-management choice. Prediction becomes decision-relative in the same way. An account centered on common semiconductor exposure calls for predictions under a semiconductor shock, not merely a forecast of continued market concentration.

The decision objective therefore shapes both the relevant representation and the prediction question. Whether competing accounts make distinguishable predictions remains open at this stage. We first need to separate two kinds of change that can occur as the decision develops.

The decision maker may revise beliefs and actions as evidence arrives. We call this the *learning loop*. The system may also respond to action: a treatment can change a patient’s outcome distribution, or market participants can react to an exploited opportunity. We call this a *responsive embedding* when action materially changes the relevant outcome distribution or structure.

Learning and responsiveness are separate properties. A decision maker can learn from measurements of a largely unchanged mechanism. A treatment can change outcomes while the mechanism governing treatment response remains stable. These distinctions matter because diagnostic prediction can inform learning in either setting, whereas intervention-induced changes in predictive signals require an action that changes the relevant system. We next establish the relation through which prediction can supply diagnostic evidence.

3. Prediction and Attribution Within the Problem

Prediction and attribution answer different questions within the same decision. The distinction between explanatory and predictive modeling is well established, including the fact that strong performance on one objective need not establish strong performance on the other [1,2].

Prediction asks what outcome to expect under a relevant condition or contemplated action. Attribution asks which components, relations, or mechanisms account for consequential differences in those outcomes. We use *attribution* in this functional sense: it can be causal, diagnostic, or risk-oriented, depending on the task and available evidence. Calling something an attribution does not establish that it is correct or empirically distinguishable from its competitors.

The portfolio example reveals two ways an attribution can affect prediction. One account emphasizes market-cap weighting; another emphasizes common economic exposure; a third emphasizes differences among the underlying businesses. These accounts may first change the question the manager asks. Identifying a shared exposure can shift attention from whether concentration persists to how large a loss that exposure could produce. We call this *attribution-guided formulation*: the account changes the prediction target, condition, or horizon.

Formulating a useful question still leaves the account to be tested. For that purpose, hold the outcome and diagnostic condition fixed. Let Y be the outcome of interest, u a common action or condition, and A_i a candidate structural account. Write the prospective distribution under that account as

$$P(Y \mid u, A_i). \quad (1)$$

The notation says that each account is asked to predict the same outcome under the same condition. For the portfolio, the comparison might concern losses under a specified factor shock, with the portfolio and horizon held common. If the accounts predict different responses, an observation under that condition can carry evidence about their relative credibility.

The meaning of u depends on the comparison. It may be an observed diagnostic condition, a contemplated action, or an implemented intervention. An intervention claim requires an intervention interpretation and the corresponding identification conditions; ordinary conditioning alone does not supply it. We retain the compact notation while specifying that meaning in each application.

The distinction between formulation and diagnostic comparison determines what the next step must accomplish. Changing the forecast target can make a decision more intelligible without testing any attribution. To establish the stronger reciprocal relation, we need accounts that imply different prospective outcomes and evidence capable of distinguishing those implications.

4. Prediction–Attribution Conjugacy

For a given decision task, prediction and attribution are conjugate when at least some decision-relevant differences among accounts imply distinguishable prospective outcomes under a common informative condition, and observations under such conditions can revise the accounts’ relative credibility. The coupling can be represented schematically as

$$A_D \leftrightarrow P_D. \quad (2)$$

Here A_D denotes the family of accounts relevant to the decision, and P_D the corresponding predictive inquiry, formalized as a mapping in Appendix A.2. The arrow represents prospective implications in one direction and diagnostic evidence in the other. This is an organizing relation between inquiries, not an algebraic identity.

The relation is reciprocal but not symmetric. An account supplies assumptions from which prospective implications follow. Observations can then reweight the accounts, but a successful prediction need not uniquely identify the structure that produced the outcome. Its diagnostic value depends on the alternatives being compared and the conditions under which evidence is obtained.

Suppose two accounts, A_1 and A_2 , are compatible with current evidence but predict different responses to the same relevant condition u :

$$P(Y \mid u, A_1) \neq P(Y \mid u, A_2). \quad (3)$$

The attributional distinction now has prospective meaning. If the condition can be observed or created, the outcome measured adequately, and the comparison protected sufficiently from selection, confounding,

and invalidating structural change, later evidence can shift support between the accounts. Bayesian model discrimination provides a neighboring formal precedent for this use of prospective outcomes [14].

A retrospective explanation may fit what has already happened while supplying no different expectation under another relevant condition. Conjugacy asks for the additional prospective comparison. We distinguish two levels of that comparison because a difference on paper may remain inaccessible in practice.

Structural conjugacy is present when decision-relevant accounts imply different prospective outcomes under at least one common informative condition. Identified conjugacy adds the empirical requirement: feasible observations or interventions must provide enough diagnostic variation to discriminate among the accounts before relevant change invalidates the comparison. Here, identification means sufficient discrimination for the task; it does not require recovering one uniquely correct causal account.

Two accounts of portfolio risk may imply different stress responses yet remain unresolved if the portfolio has never encountered, and cannot feasibly be tested under, a sufficiently diagnostic stress. A one-off policy or investment choice can similarly depend on competing accounts while offering no informative comparison before action is required. The decision problem remains real even when identified conjugacy is unavailable.

Table 1 brings together the steps developed so far. Each step answers a question left open by the preceding one: an account identifies relevant structure, formulation selects the next prediction question, a common-condition comparison exposes prospective differences, and diagnostic observations can revise the accounts.

Table 1. Four related layers in the framework

Layer	Question	Role
Attribution	What structure matters to the decision?	Supplies a candidate account.
Attribution-guided formulation	What should be predicted next?	Changes the target, condition, or horizon.
Attribution-conditioned prediction	What does each account predict for the same outcome under the same condition?	Exposes prospective differences.
Prediction–attribution conjugacy	Can observations of those differences revise the accounts?	Establishes the reciprocal diagnostic relation, where feasible evidence permits it.

The conjugate core can operate within a learning loop even when the relevant mechanism is stable. When the pair also guides action that changes the system, we must follow what that action does to later predictions and evidence. The next section develops these consequences in that order.

5. What the Conjugacy View Helps Us See

5.1 Attribution can change the prediction problem

The first consequence arises before diagnostic testing. The portfolio manager’s account of common exposure redirects the prediction toward losses under a factor shock. This changes what must be forecast, not simply the accuracy of a forecast whose target was already fixed.

Causal-prediction research similarly examines predictions under hypothetical interventions [6,7]. Its relevance here is that a contemplated action can require a different prediction question from the one answered by a historical observational model. We therefore assess attribution-guided formulation by whether it identifies the outcome, condition, or horizon useful for the decision. A separate test is needed to assess the credibility of the account that guided it.

5.2 Predictive outcomes can discipline attribution

That separate test begins when competing accounts make different claims under a common diagnostic condition. For the manager, a shared-factor shock could distinguish an account of common vulnerability from one centered on largely independent business risks. Evidence from a condition under which both accounts predict much the same result would have less discriminating value.

Prospective predictions thus give attribution empirical exposure. We use *discipline* to mean that observations can change the evidentiary balance among accounts when the comparison is sufficiently diagnostic and credible. The amount learned depends on outcome variability, measurement, and control over competing explanations. Ordinary forecast success supplies no general guarantee of unique causal identification.

So far, this relation concerns what the decision maker can learn. Once the resulting understanding guides action, a further question arises: does the action change the system against which the next prediction will be tested?

5.3 Acting on the prediction–attribution pair can change the system

In a responsive embedding, the prediction–attribution pair informs an action that materially changes the relevant outcome distribution or structure. Using A_t and P_t for the accounts and predictions currently informing action, and u_t for the selected action, the sequence is schematically

$$A_t \leftrightarrow P_t \rightarrow u_t \rightarrow S_{D,t+1}. \quad (4)$$

The final arrow marks the contribution of action to the next decision-relevant system; external events can contribute as well. Appendix A.7 separates the action rule from the transition. A treatment may change outcomes mechanically, while a trading strategy may elicit reactions from other participants. A stable mechanism can therefore support a responsive outcome distribution; structural change is one possible response, rather than a requirement for every response.

Dual-control research examines settings in which action both controls a system and generates information about its uncertainty [8]. Performative prediction formalizes settings in which prediction-supported decisions alter future outcome distributions [9]. These neighboring accounts help locate the surrounding

dynamics. The conjugacy view directs attention to how subsequent observations bear on the account and predictor that supported the action.

That distinction becomes useful when an old predictive relationship weakens. We need to know whether the account was mistaken, the structure changed for another reason, or action altered a condition on which the old signal depended.

5.4 Successful intervention can weaken an old predictive signal

Consider a variable X that contributes to outcome Y and whose variation helps distinguish higher-risk from lower-risk cases. An attribution identifies X as actionable. If intervention reduces the relevant variation in X , the signal may have less remaining ability to distinguish risks even when the mechanism relating X to Y stays intact. If intervention changes that relationship itself, forecasts based on the earlier relation may also become miscalibrated.

These are different forms of deterioration. Loss of discrimination means the signal separates outcomes less well. Loss of calibration means predicted probabilities no longer agree adequately with outcome frequencies under the conditions assessed. A successful intervention can reduce the first without producing the second. For example, removing a source of risk may leave everyone with similarly low risk, correctly predicted by a model using their updated circumstances.

The consequence is conditional. A uniform change that preserves informative differences, or an intervention that leaves the relevant relationship unchanged, need not weaken the signal. Nor does a reduction in signal information determine the direction of every accuracy or loss measure. The metric and the information supplied to the predictor must be specified.

Monitoring can nevertheless use the intervention's expected consequences to distinguish explanations. A changed outcome pattern confined to appropriately treated cases is different evidence from similar change among comparable untreated cases. The diagnostic task is to relate what changed to what the competing accounts predicted, while checking implementation and other explanations. Appendix A.12 supplies a numerical comparison and separates an intervention's benefit from changes in discrimination and calibration.

5.5 Diagnostic learning can be overtaken by structural change

A further problem is timing: evidence may accumulate too slowly to distinguish accounts before relevant change invalidates the comparison. We call this the identification race.

The important comparison is between the time needed to obtain task-adequate diagnostic evidence and the time over which the diagnostic relationship remains usable. If evidence arrives much sooner, understanding may accumulate across a relatively stable comparison. If the timescales are similar, accounts may remain provisional. If relevant change comes first, resolving the earlier distinction may no longer answer the present decision.

We denote the corresponding conceptual rates by λ_I for diagnostic identification and λ_R for invalidating structural change. Appendix A.11 expresses them as inverse characteristic times, so both have the same temporal units. They are heuristic descriptors of a specified comparison, not universal measured

constants. Exploratory actions can accelerate learning, while intervention can also affect the pace of change.

This comparison directs attention to the timing of evidence without turning every difficulty into a timing problem. Noise, measurement limits, and lack of experimental access can prevent identification even in a stable system. We next use contrasting cases to show where diagnostic access and responsiveness matter separately.

6. The Structure Across Different Problems

The cases illustrate how the same questions operate under different decision conditions: which accounts matter, where their predictions diverge, whether that difference can be observed, and what action changes. They are demonstrations of the framework's use and boundaries, rather than empirical validation of the framework as a whole. Appendix B organizes the cases by diagnostic access and responsiveness.

6.1 Portfolio concentration risk

The opening case is strongest as an illustration of decision-relative formulation. Accounts centered on portfolio weights, shared factors, and distinct business risks can lead to different stress questions and hedges. Once the outcome and stress are held common, those accounts may also have structurally distinguishable implications.

The empirical comparison can remain weak because diagnostic stresses are rare or model-based scenarios depend on assumptions still to be tested. A stress calculation displays implications of an account; it does not itself establish that the account is correct. Responsiveness also depends on the chosen system. A small hedge may leave the wider market largely unchanged while materially changing the portfolio's outcome distribution. The relevant boundary is therefore the decision-relevant system, not finance as a whole.

6.2 Making money in adaptive financial markets

An investor seeking returns faces a different reason to understand the same market. A return opportunity may be attributed to temporary mispricing, persistent risk compensation, institutional constraints, behavioral regularities, or informational advantage. These accounts imply different expectations about durability and the consequences of exploitation.

The Adaptive Markets Hypothesis emphasizes competition and adaptation in changing financial environments [13]. The conjugacy view focuses on the diagnostic relation within such an environment: an account supports prospective expectations, observed returns can revise that account, and successful exploitation may alter the opportunity itself.

A deteriorating strategy consequently invites several explanations. Its initial pattern may have been spurious, the relevant market structure may have changed independently, or exploitation may have weakened the signal. The last explanation is most directly applicable when the opportunity is competitively exploitable and use affects the relationship generating it. Other sources of excess return may persist. Distinguishing these possibilities requires evidence about the opportunity and its response, not simply a record of declining returns.

6.3 Adaptive clinical trials

Clinical trials offer a more controlled route to diagnostic evidence. Competing accounts of treatment response can imply different outcomes under prospectively specified treatments, and observations can revise those accounts [11]. An adaptive design may change allocation or stopping rules as evidence accumulates.

This adaptation belongs to the learning and decision process. Treatment can also make the decision-relevant system responsive by changing participants' outcome distributions, even while the biological response mechanism remains stable. A trial therefore need not involve changing mechanisms to combine identified conjugacy with responsiveness. A non-interventional diagnostic measurement of a stable mechanism would supply a different, potentially non-responsive comparison.

The useful contrast with the financial cases is diagnostic access. Randomization or other suitably designed comparisons can separate treatment responses under specified conditions. Whether the evidence distinguishes the candidate accounts still depends on the design and their prospective implications. Adaptive allocation itself does not guarantee better identification and can involve efficiency or power trade-offs.

6.4 Public policy during COVID-19

COVID-19 policy illustrates a less controlled setting. Governments acted while transmission, behavior, treatment capacity, immunity, variants, mobility, economic conditions, and public responses changed. OECD's synthesis of early government evaluations emphasizes learning what was working, what was not, what could work, and for whom [12].

Those changes create a problem for comparing prospective implications. An outcome observed after a policy may reflect the policy, changing behavior, a different epidemiological condition, or their interaction. Evidence capable of distinguishing accounts can arrive after the conditions underlying the earlier comparison have changed.

This illustrates a possible identification race. It does not estimate λ_I or λ_R from the OECD source. Applying the comparison empirically would require a specified policy question, competing accounts, and evidence about when their diagnostic contrast remained usable.

6.5 A boundary case: carrying an umbrella

Suppose a forecast gives a high probability of rain tomorrow and the decision is simply whether to carry an umbrella. With the relevant costs and inconvenience understood, the forecast may supply enough information for that choice. A more detailed account of atmospheric mechanisms may be scientifically valuable without changing the recommendation.

Carrying the umbrella also leaves the weather unchanged and supplies little diagnostic evidence about atmospheric dynamics. The case therefore lacks the decision-relevant attributional distinction needed for the present framework. Its value as a boundary case is to keep consequential prediction separate from a decision that also needs diagnostic structural understanding.

The cases together show why a domain name cannot determine classification. Finance can offer weak diagnostic access, a trial can offer strong access with a stable mechanism, and public policy can involve both intervention and rapid change. We now relate these distinctions to the research traditions that develop their component parts.

7. Relation to Existing Work

The framework brings together questions studied from several directions. We begin with prediction and explanation because their distinct purposes establish why a good forecast need not supply the structural understanding required for action. Shmueli distinguishes explanatory and predictive modeling [1], while Hofman and colleagues examine how explanation and prediction can complement one another [2]. Our organizing question concerns their decision-relevant relation: when can an account generate diagnostic predictions and those predictions supply evidence about the account?

The decision determines whether attribution is needed. Kleinberg and colleagues identify policy problems in which predictive inference has an independent role [3]. Causal-prediction research addresses forecasts under hypothetical interventions [6], while invariant prediction uses predictive stability across environments as information about causal structure [7]. These approaches help distinguish a useful prediction target from a comparison capable of testing structural claims.

Once competing accounts have prospective implications, experimental design becomes relevant. Bayesian model discrimination provides methods for selecting observations or interventions that separate competing models [14]. Decision-focused learning addresses the related distinction between predictive accuracy and decision quality [15]. These traditions give formal content to questions about which evidence to obtain and which predictive errors matter for action.

Action then connects the diagnostic comparison to the wider system. Adaptive control and system identification study cycles of modeling, prediction, action, observation, and update. Dual control makes action’s informational role explicit [8], while performative prediction examines how prediction-supported decisions change future distributions [9]. Our separation of the conjugate core from responsiveness helps locate what comes from diagnostic inference and what comes from action-induced change.

Several works also use terminology close to ours. Efron distinguishes prediction, estimation, and attribution in statistical inference [4]. Rayl and Sinha integrate attribution-based and pure-prediction approaches and document trade-offs between them [5]. Tikhonov and DeDeo examine prediction, explanation, and control as related but distinct abilities in dynamic environments [10]. These comparisons support keeping the relation task-specific rather than assuming a universal trade-off.

Taken together, these literatures provide many ingredients of the present account. We organize them around decision-relevant attribution, diagnostic prediction, and what subsequent action changes. That organization is useful insofar as it helps distinguish an unresolved account from a failed predictor, and either from a signal changed by successful intervention. The next section follows those distinctions into monitoring and continued learning.

8. Implications and Limits

8.1 Monitor the account and predictor together

A monitor that records acceptable average predictive performance under routine conditions may leave the underlying attribution unresolved. If competing accounts differ only under stress, treatment, a subgroup condition, or a shifted environment, routine outcomes supply little evidence about the distinction that matters for action.

Monitoring the account and predictor together therefore means asking where their predictions diverge and preserving or creating credible comparisons there. Depending on the task, these might involve holdouts, randomized exposure where feasible, observed stress events, or subgroup and shifted-environment comparisons. Model-based stress scenarios can clarify prospective differences while leaving their empirical validity to be assessed.

Attributional credibility is assessed through such diagnostic predictions, interventions, and other evidence. Predictive performance must likewise be attached to an outcome, condition, and metric. A stable account can coexist with reduced discrimination after a successful intervention, while good routine accuracy can coexist with uncertainty about the account. The practical question is whether the prediction remains useful for the reason assumed, and whether that reason has been examined where competing accounts could disagree.

8.2 Understanding can improve without becoming final

Diagnostic evidence can narrow the range of credible accounts while the relevant system continues to change. A decision maker may understand an earlier condition more clearly without yet having a settled account of the condition now faced. The identification race helps distinguish slow learning about a stable comparison from learning whose object changes before uncertainty is resolved.

A related extension concerns life choices. A career or education decision can be reasonable given the information and self-understanding available at one time, while the decision itself changes later opportunities or preferences. A later choice may then reflect both learning and changed circumstances. This is a possible application of the distinction, rather than a result established for life decisions by the present analysis.

8.3 Diagnostic questions in the AI landscape

The AI landscape offers another domain for applying these questions. Suppose a team attributes a system's success on current-information tasks to information recovered from training, while another account emphasizes retrieved sources. Performance under routine conditions may fit both accounts. Controlled changes in source availability or currency could help separate their prospective implications, provided other relevant conditions remain sufficiently comparable.

A deployment change can then alter the outcome distribution faced by the earlier performance model. Following the account, its diagnostic prediction, and the changed configuration would help interpret the

resulting performance. This example illustrates an application of the framework's questions; evaluating their usefulness requires observations from specified systems and tasks.

8.4 Apply the distinctions at the level of the problem

The broad setting is consequential choice under structural uncertainty, subsequent observation, and possible revision. Within that setting, structural conjugacy asks whether relevant accounts have different prospective implications under a common condition. Identified conjugacy asks whether feasible evidence can distinguish those implications before invalidating change. Responsiveness separately asks what the decision maker's action changes.

These tests keep the framework's scope tied to its purpose. A forecast may be sufficient for an umbrella decision; a policy choice may require structural understanding while offering no timely diagnostic comparison; a trial may supply a credible comparison without a changing biological mechanism. Prediction, feedback, or adaptation alone does not establish every property.

Appendix C develops exploratory directions concerning attainable combinations of predictive and attributional quality, information limits, identification timing, and continuous-time dynamics. They extend questions raised by the analysis while remaining separate from its central diagnostic claim.

9. Conclusion

The portfolio manager's opening forecast becomes useful for action only after the relevant question is clarified. Persistent concentration may call for different responses depending on whether the concern lies in portfolio weights, shared exposures, or business-specific risks. An attribution helps determine what to predict next. Once the outcome and condition are held common, different prospective implications can also create an opportunity to test the competing accounts.

We call that reciprocal diagnostic relation prediction–attribution conjugacy. Its practical value depends on whether the distinction matters to the decision and whether a feasible comparison provides evidence about it. A prospective difference is a reason to seek diagnostic evidence, not proof that identification is available.

Using the pair adds a further relationship. Action can change the outcome distribution or structure encountered next, so monitoring must follow both what the earlier account predicted and what the intervention was intended to change. A weaker signal may record an earlier mistake, independent change, or successful reduction of the variation supplying the signal. Distinguishing discrimination from calibration prevents those possibilities from being collapsed into one verdict about accuracy.

The same need for diagnostic evidence creates a timing constraint. Understanding can accumulate when useful comparisons remain available long enough to resolve relevant uncertainty. It may remain provisional when evidence arrives slowly or the contrast changes before the decision can use it. Responsiveness and diagnostic access therefore matter separately and together.

This organizing view connects established methods to a question the decision maker can use: what understanding supports the prediction, where could competing accounts disagree, and what happens to the

account and predictive signal once we act? Following those relationships helps identify the next evidence to obtain, the control to test, and the interpretation warranted by a changing result.

AI-Use Disclosure

AI tools were used in the development of this working paper for exploratory discussion, literature discovery, drafting assistance, and editorial revision. The author selected the research question, developed and directed the argument, evaluated the resulting claims and structure, and takes responsibility for the final content.

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Appendix A. Formal Definitions and Minimal Model

The notation makes the dependencies used in the main discussion explicit. We define the decision-relevant objects first, then distinguish prospective comparison, diagnostic learning, and responsiveness. The expressions are definitions or conceptual schemas unless a stated example supplies a probability model. They do not establish empirical effects by themselves.

A.1 Decision environment

Let S denote the broader system and D a decision objective. Let π_D be the representation rule selecting or organizing the features relevant to that objective. Define

$$S_D = \pi_D(S). \quad (\text{A1})$$

Let U_D be the available action set, $u \in U_D$ an action, and Y a consequential outcome assessed over a specified horizon. The decision problem is organized around (D, S_D, U_D, Y) . A diagnostic condition may instead be observed rather than chosen; we specify which role it has in an application. The representation can be incomplete or uncertain, and its adequacy is part of the inquiry.

A.2 Prediction

A predictive mapping associates the decision-relevant representation and an action or condition with a prospective outcome distribution:

$$P_D: (S_D, u) \mapsto P(Y \mid S_D, u). \quad (\text{A2})$$

The output is a probability distribution, rather than a structural explanation. Under a candidate account A_i , we write $P(Y \mid u, A_i)$ for the account’s prospective distribution, suppressing the fixed decision specification and horizon. When u is an intervention, this distribution requires intervention semantics and appropriate assumptions; observational conditioning alone is insufficient.

A.3 Attribution

Let A_D denote a family of attributional hypotheses relevant to the decision. Each A_i is a candidate account of which components, relations, or mechanisms within S_D matter to consequential differences in outcomes. An account may be causal, diagnostic, or risk-oriented under the task's evidentiary standards.

An attribution's existence does not imply that its prospective implications are distinct or identifiable. Those properties are assessed through the comparison developed next.

A.4 Attributional equivalence

Fix the objective, available information, constraints, action set, and decision criterion. Write

$$A_i \sim_D A_j \tag{A3}$$

when the two accounts imply the same action recommendations under those fixed conditions. The equivalence class $[A]_D$ groups accounts that can differ scientifically or descriptively while remaining indistinguishable for that decision specification. Changing the objective, information, or criterion can change the equivalence classes. This decision-relative equivalence should be distinguished from fitting the same observed data.

A.5 Predictive distinguishability

For a fixed outcome and horizon, two accounts are predictively distinguishable if at least one common relevant action or diagnostic condition yields

$$P(Y \mid u, A_i) \neq P(Y \mid u, A_j). \tag{A4}$$

The distributions must concern the same outcome under the same condition. Comparing different targets or different actions would mix formulation with diagnostic distinguishability. The distributional inequality establishes a prospective difference; whether that difference can be measured adequately is a further empirical question.

A.6 Prediction–attribution conjugacy

Structural conjugacy exists when at least some decision-relevant distinctions among accounts have different prospective implications under a common informative condition. Identified conjugacy adds feasible observations or interventions that discriminate sufficiently among those accounts before relevant change invalidates the comparison.

The schematic relation $A_D \leftrightarrow P_D$ represents prospective implication and diagnostic evidence. It describes a coupling of inquiries, not an inverse operation. Discrimination can revise relative credibility without identifying one uniquely correct causal account.

A.7 Learning loop and responsive embedding

Let t index decision rounds. Let A_t denote the account or collection of weighted accounts currently used, and P_t the associated predictive assessment. Let h_D be the action rule under objective D . Then

$$u_t = h_D(A_t, P_t). \quad (\text{A5})$$

Let T denote the transition of the decision-relevant system and ε_{t+1} the external influences entering the next round:

$$S_{D,t+1} = T(S_{D,t}, u_t, \varepsilon_{t+1}). \quad (\text{A6})$$

These expressions separate action selection from system transition. The decision specification and available information are held implicit in h_D ; the transition is a conceptual schema rather than an estimated causal model. Observations from the next round can contribute to revision from (A_t, P_t) to (A_{t+1}, P_{t+1}) , which is the learning loop.

Responsiveness requires action to materially change the relevant outcome distribution or structural relationships. It can arise through a stable mechanism, as when treatment changes outcomes, or through a changed mechanism or strategic response. Learning can occur without responsiveness if the diagnostic act leaves the relevant system materially unchanged.

A.8 Predictive sufficiency and decision sufficiency

Predictive sufficiency concerns a specified target and standard of performance. Decision sufficiency concerns the information needed to choose appropriately under the objective and constraints. Predictive sufficiency for one target need not supply decision sufficiency when action depends on additional outcomes, conditions, or structural distinctions that the target does not encode.

The opening concentration forecast illustrates the dependency: predicting concentration can be useful while leaving the portfolio's loss exposure and appropriate response unresolved.

A.9 Success-induced signal attenuation

Let X be a measured signal representing a relevant component or relationship in S_D , and let Y be the outcome it helps predict. If action changes the distribution of X or the relationship between X and Y , the predictive information in that signal may decline. One possible description is

$$I_{\text{post}}(X; Y) < I_{\text{pre}}(X; Y), \quad (\text{A7})$$

where $I(X; Y)$ is mutual information under the respective joint distribution, and “pre” and “post” refer to populations and periods specified for the comparison. This is a possible inequality, not a consequence of every intervention or uniform shift.

Reduced mutual information does not determine the change in every forecast-performance metric. A signal can lose discrimination while probability forecasts using updated inputs remain calibrated. If action changes the conditional outcome relation, a predictor retaining the old relation may instead become miscalibrated. Appendix A.12 makes the first distinction explicit.

A.10 Pair-level validity

Let $V_P(t)$ describe predictive performance for the specified target, conditions, and metric at round t . Let $V_A(t)$ describe the evidentiary credibility of the account relative to its competitors. These are schematic summaries; assessing attribution requires evidence under conditions that separate the accounts.

Stable V_P under non-diagnostic conditions need not establish stable V_A . Conversely, a stable account need not preserve every predictive-performance measure after the distribution of an informative variable changes. Monitoring must therefore identify the relevant contrast and metric, rather than treat either summary as a universal measure of validity.

A.11 The identification race

For a specified comparison, let τ_I be a characteristic time needed to obtain task-adequate diagnostic evidence, and τ_R the characteristic time over which relevant change would invalidate that comparison. When both times are positive and finite, define

$$\lambda_I = \tau_I^{-1}, \quad \lambda_R = \tau_R^{-1}, \quad \rho = \frac{\lambda_I}{\lambda_R} = \frac{\tau_R}{\tau_I}. \quad (\text{A8})$$

Both rates have units of inverse time, and ρ is dimensionless. A large ratio describes evidence arriving faster than invalidating change; a small ratio describes the converse. These descriptors depend on the chosen diagnostic contrast and evidentiary threshold. They are heuristic, not independent measured constants or sufficient conditions for successful learning. In a stable comparison without a finite invalidation time, the limiting interpretation should be used rather than the finite ratio.

A.12 A diagnostic toy model

We first specify two observationally equivalent accounts, then distinguish them through an intervention. All values below are hypothetical. Let X and Y be binary variables. Before intervention, half the population has $X = 1$, and both accounts imply

$$P(Y = 1 \mid X = 1) = 0.60, \quad P(Y = 1 \mid X = 0) = 0.20. \quad (\text{A9})$$

Under A_1 , X contributes directly to the event risk: setting $X = 1$ gives risk 0.60, and setting $X = 0$ gives risk 0.20. Under A_2 , an unobserved binary driver Z determines risk in the same way, while $X = Z$ before intervention. Thus both accounts fit the same observational probabilities, but they differ about whether changing X alone changes risk.

Among participants initially having $X = 1$, randomly assign an intervention that sets $X = 0$ without directly changing Z , retaining a comparable untreated group. Assume the intervention is implemented as specified, with no spillover between groups and no other change invalidating the comparison. The accounts now imply different treated risks:

$$P(Y = 1 \mid u_{\text{treat}}, A_1) = 0.20, \quad P(Y = 1 \mid u_{\text{treat}}, A_2) = 0.60. \quad (\text{A10})$$

Both predict an untreated risk of 0.60 for this initially exposed subgroup. If observed treated and untreated rates are 0.25 and 0.60, respectively, the pattern favors A_1 under the stated conditions. The

strength of that evidence still depends on sample size, variability, measurement, and implementation; the illustrative rates alone establish no statistical significance or unique causal identification.

A separate comparison shows why signal attenuation is not necessarily forecast miscalibration. Suppose A_1 is the operative mechanism and the intervention sets $X = 0$ throughout the population. Before intervention, X varies and carries information about Y . Afterward, X is constant, so $I_{\text{post}}(X; Y) = 0$, while event risk is 0.20. The causal account remains intact and the intervention lowers risk for previously exposed participants, but current X can no longer distinguish higher-risk from lower-risk cases.

A forecast using current X and the unchanged conditional relationship correctly assigns risk 0.20 to everyone. A frozen forecast that continues to use the old exposure status would instead assign 0.60 to previously exposed participants and overpredict their risk. These are distinct monitoring outcomes: loss of discrimination from compressed variation, and miscalibration from stale predictor inputs.

The broader three-way diagnosis still requires evidence. An invalidating change affecting treated and untreated groups could undermine the earlier comparison; failure to deliver the intervention could mimic an ineffective mechanism. Following the predicted contrast, implementation, and comparison group is what connects a performance change to an explanation.

Appendix B. Extended Cases and Classification

The classification separates two questions: can feasible evidence distinguish the relevant accounts, and does the action materially change the decision-relevant outcome distribution or structure? Structural conjugacy may be present while diagnostic access is insufficient. The unit of classification is the specified decision problem, not its domain.

Table B1. Identified conjugacy and responsive embedding

Diagnostic status	Responsive embedding absent	Responsive embedding present
Identified conjugacy absent	A one-off structural choice whose competing accounts matter, but whose diagnostic condition cannot be observed before the choice; the chosen action leaves the specified system materially unchanged.	A policy or platform choice with structural uncertainty and an action-induced change, but no feasible diagnostic contrast before the comparison becomes invalid.
Identified conjugacy present	Non-invasive diagnosis of a stationary reliability mechanism: accounts predict distinguishable measurements, while the diagnostic act leaves the mechanism and relevant outcomes materially unchanged.	A treatment trial with diagnostic comparisons and treatment-induced changes in outcome risk; adaptive finance or performative credit where diagnostic evidence is available and action changes the relevant opportunity or population.

Responsiveness need not involve a changing mechanism. Treatment can change an outcome distribution through a stable mechanism, and a small portfolio action can change portfolio risk without altering the wider market. Conversely, merely observing a changing system does not establish that the decision maker's action caused that change.

The umbrella problem remains an out-of-class comparison when atmospheric attribution does not change the action recommendation. A reactive recommender can likewise fall outside the problem class when attribution is operationally irrelevant, even though the system responds to decisions. Adaptation or feedback alone is therefore insufficient to establish the conjugate diagnostic relation.

B.1 What the extended cases add

The matrix separates learning, diagnostic access, and response. The same domain can contain problems with or without identified conjugacy, and with or without a responsive embedding, depending on the objective, system boundary, action, and available evidence. A useful application states those conditions before assigning the problem to a cell.

Appendix C. Further Conjectures and Open Directions

The following directions extend the questions developed in the paper. They are proposals for further formalization or investigation, rather than established general results.

C.1 A moving attainable frontier

Let $\mathcal{I}_t$ denote the information available at time t , q_A a task-specific measure of attributional quality, and q_P a measure of predictive quality for the chosen target. Define the attainable set schematically as

$$\mathcal{F}_t(D, \mathcal{I}_t) = \{(q_A, q_P) : (q_A, q_P) \text{ is attainable}\}. \quad (\text{C1})$$

The quality criteria and resource constraints would need to be specified in an application. Better attribution may improve prediction, intervention may reduce an old signal's discrimination, and the two qualities may change largely independently elsewhere. The open question is whether particular classes of conjugate problems have useful common structure in the boundary and movement of this set.

C.2 Information and identifiability limits

If decision-relevant states have the same observable implications under all comparisons currently available but require different actions, existing evidence cannot recover that distinction alone. New measurement, variation, or intervention may be needed. The open direction is to characterize which decision-relevant distinctions are recoverable under a specified information and access constraint.

C.3 The identification race as a dynamic limit

The timing comparison in Section 5.5 suggests studying whether task-adequate evidence can accumulate before a diagnostic contrast becomes invalid. A formal model would need to specify the evidence process, the threshold for useful discrimination, the relevant structural changes, and how exploratory or consequential actions affect both. Those dependencies determine whether the ratio in Appendix A.11 is a useful summary.

C.4 Continuous-time dynamics

As the interval between updates becomes small, attribution, prediction, action, and system evolution may become closely co-evolving rather than cleanly sequential. A future formalization could examine coupled processes $A(t)$, $P(t)$, $u(t)$, and $S_D(t)$. The action is written $u(t)$ to distinguish it from the fixed decision

objective *D*. An appropriate dynamic representation would depend on the particular learning and response mechanisms.